

Diuretics Pharmacology is one lecture in two parts. The learning objectives are the same for both. After completing the preparation materials, students should be able to:

1. **Sites of action:** Map each diuretic class to its site of action in the nephron and predict the downstream effects.
2. **Mechanisms:** Explain how inhibition of specific renal transporters alters sodium handling, electrochemical gradients and downstream nephron function to produce diuresis.
3. **Effects:**
 - Correlate each diuretics class with their urinary electrolyte patterns.
 - Predict acid-base disturbances based on mechanisms.
 - Explain diuretic effects on tubuloglomerular feedback, renal hemodynamics, GFR.
4. **Compensatory responses:** Relate diuretic actions to compensatory physiological responses (e.g., SNS, RAAS activation, distal nephron reabsorption) and explain their role in diuretic resistance.
5. **Therapeutic uses:** Select the appropriate diuretic based on mechanism, patient factors, and clinical scenario.
6. **Clinical considerations:**
 - Explain how the pharmacokinetic properties influence onset, efficacy, and clinical use of the diuretic agents.
 - Predict the adverse effects and contraindications based on diuretic mechanism and patient factors.
 - Predict the impact of NSAIDs on diuretic response by describing the role of the renal prostaglandins on the actions and effects of the diuretic agents.
 - Predict drug interactions based on the mechanism of action, pharmacokinetic properties, or pharmacogenetic variabilities of the individual drugs or drug classes

Key points: Thiazide diuretics

- Thiazide-type and thiazide-like diuretics (broadly categorized as thiazide diuretics) act on the sodium chloride cotransporter (NCC) in the distal convoluted tubule where ~5-10% of sodium is reabsorbed. Blocking NCC leads to increased sodium, chloride, and water excretion.
- Thiazide diuretics cause mild diuresis, which reduces plasma volume and decreases cardiac output, resulting in lower blood pressure.
- With continued therapy, the plasma volume returns to pretreatment levels (a desirable effect), with long-term antihypertensive effects.
- Thiazide diuretics are recommended first-line agents for the management of primary hypertension and are useful in the treatment of Liddle syndrome and vasopressin resistance (nephrogenic diabetes insipidus). (vasopressin = antidiuretic hormone)
- Thiazide diuretics cause potassium-wasting effect, mild hypokalemic metabolic alkalosis, and hyponatremia and are associated with increased serum glucose, lipids, and uric acid.
- Thiazides increase urinary magnesium loss by impairing the apical magnesium channel (TRPM6) in the DCT.
- Students should be aware of specific adverse effects and drug interactions.

Key points: Potassium-sparing diuretics

- Epithelial sodium channel (ENaC) inhibitors and mineralocorticoid receptor antagonists (MRAs) are potassium-sparing diuretics that inhibit sodium reabsorption in the principal cells located in the late DCT, connecting tubule, and collecting ducts. Sodium reabsorption in the principal cells is tightly regulated by aldosterone (aldosterone-sensitive sites).
- They are considered weak diuretics because they block reabsorption of a small percentage of filtered sodium (~2-5%).
- They are known as the potassium-sparing diuretics because they prevent loss of potassium in the urine. There is a risk of hyperkalemia especially when administered with other potassium-retaining drugs.
- Potassium-sparing diuretics are typically given with another diuretic for additive effect and to offset potassium loss.
- ENaC inhibitors, amiloride and triamterene, reduce the passive apical entry of sodium ions, which reduces the lumen-negative transepithelial potential.
- The MRAs, spironolactone, eplerenone, and finerenone, block the actions of aldosterone, an adrenal hormone that promotes sodium reabsorption and potassium excretion.
- Students should be aware of specific adverse effects and drug interactions.

Key points: Osmotic diuretics

- **Kidney:** Osmotic diuretics are freely filtered at the glomerulus, not reabsorbed, and are pharmacologically inert (no receptor targets). They stay in the tubular lumen, increasing the osmolarity of the luminal fluid. The greatest effect is in the water permeable segments – the proximal tubule (major effect) and the descending limb of the loop of Henle.
- The increased luminal flow reduces passive Na⁺ reabsorption and substantially increases urine output, producing rapid diuresis.
- **Plasma:** Osmotic diuretics, such as mannitol, stay in extracellular fluid. Plasma osmolality rises, causing water to shift out of cells into the plasma.
- **Therapeutic uses:** By increasing the osmotic pressure of plasma, osmotic diuretics extract water from the eye and brain. Osmotic diuretics are used to control intraocular pressure during acute attacks of glaucoma and for short-term reductions in intraocular pressure both preoperatively and postoperatively in patients who require ocular surgery. Also, mannitol is used to reduce cerebral edema and brain mass before and after neurosurgery and to control intracranial pressure in patients with traumatic brain injury.
- **Adverse effects:** Osmotic diuretics can cause pulmonary edema or worsen heart failure.

Pharmacokinetics of selected thiazide diuretics

PK Takeaway: Well absorbed, bioavailability generally good, moderate protein binding, most are not metabolized, enter nephron via filtration and OAT secretion, excreted as intact drug, half-lives vary

Drug	Oral Bioavailability	Onset of diuresis Duration of action	Route of elimination	t _{1/2} elimination (approximate)
HCTZ hydrochlorothiazide	70%	2 hours 6-12 hours ←	Urine	6-15 hours
Chlorthalidone	65%	2-6 hours 24-72 hour ↗	Urine 50-65% Hepatic metabolism and fecal excretion	40-60 hours ↗
Chlorthalidone Takeaway: Very long t _{1/2} , long duration of action provides smoother 24-hour blood pressure control. Higher risk of hypokalemia than with HCTZ				

Plasma protein binding varies considerably among thiazide diuretics. Hydrochlorothiazide, chlorothiazide, and chlorthalidone are approximately 70-75% plasma protein bound at therapeutic doses.

t_{1/2} adults, normal renal function

Chlorothiazide is the only intravenous thiazide diuretic.

Indapamide is extensively metabolized in the liver and has a long half-life.

Summary of Changes in Urinary Electrolyte Patterns and Body pH in Response to CA Inhibitors, Loop Diuretics, and Thiazide Diuretics

	Urinary Electrolytes			
Diuretic Class	Na ⁺	HCO ₃ ⁻	K ⁺	Body pH
Carbonic anhydrase inhibitors	+	+++	+	↓
Loop diuretics	+++++	0	+++	↑
Thiazides	++	+	++	↑
Loop diuretics plus thiazides	+++++	+	+++++	↑

+ , increase; − , decrease; 0 , no change; ↓ , acidosis; ↑ , alkalosis

RECAP: Effects of Thiazide Diuretics

↑ Na⁺ excretion ~5-10% of filtered sodium; water follows → moderate natriuresis / diuresis

K⁺ loss/hypokalemia, increased H⁺ secretion, ECF volume contraction / RAAS activation → hypokalemic metabolic alkalosis. Hypochloremia and bicarbonate retention sustain metabolic alkalosis.

Thiazide diuretics increase Ca²⁺ reabsorption and moderate Mg²⁺ loss.

Thiazide diuretics impair the kidney's ability to excrete dilute urine, which impairs excretion of free water. Thiazides do NOT impair the kidney's ability to concentrate urine, which can lead to hyponatremia.

Thiazides can produce hyperglycemia and insulin resistance, mild dyslipidemia, and increased lithium levels.

Thiazide Diuretics: Clinical Applications

Hypertension
Thiazide diuretics are first-line:
Effective, consistent BP lowering effect with stronger outcome data than several other classes of antihypertensive agents.

- Acutely lower BP by reducing ECF and cardiac output.
 - Sustained effect is due to chronic sodium depletion that decreases vascular smooth muscle responsiveness to vasoconstrictors → persistent reduction in peripheral vascular resistance.
- Hydrochlorothiazide is most widely used – in many fixed-combo products. Chlorthalidone has a much longer t_{1/2}.
- Comparative efficacy:** Large-scale observational studies and randomized clinical trials have shown that both drugs reduce blood pressure and decrease the risk of heart attacks, stroke, and heart failure at similar rates (Ishani, Cushman, et.al.; 2022).

Congestive heart failure

Liver cirrhosis (ascites)

Nephrotic syndrome

Chronic renal failure

Acute glomerulonephritis

Edematous states

Sequential nephron blockade:

Thiazide combined with a loop diuretic for synergy.

Thiazide Diuretics: Clinical Applications

Vasopressin resistance (AVP-R), including lithium-induced AVP-R (nephrogenic diabetes insipidus)

TZD + low sodium, low protein diet + NSAID

In AVP-R, the collecting duct cannot respond to ADH.

NCC blockade by thiazide diuretics shifts Na^+ reabsorption upstream.

Mechanism:

Thiazide-induced mild volume contraction $\rightarrow$ RAAS activation and $\downarrow$ GFR $\rightarrow$ Na^+ and water reabsorption in proximal tubule $\rightarrow$ substantial drop in filtrate reaching ADH-sensitive sites in collecting ducts $\rightarrow$ overall urine output decreases by up to 30-50%.

Result: overall conservation of water

Nephrolithiasis due to idiopathic hypercalciuria

Thiazides reduce urinary calcium and may reduce kidney stone recurrence in selected hypercalciuric patients.

- Thiazides decrease Ca^{2+} excretion, which lowers Ca^{2+} concentration in the urine, which decreases supersaturation of calcium salts (calcium oxalate and calcium phosphate and crystal formation.

Preventive therapy usually consists of a combination of increased fluid intake, dietary modification, weight loss, and drug therapy.

Thiazide Diuretics: Adverse Effects (most are dose-dependent)

ECF volume depletion, mild	Hyperuricemia / gout ³
Hypotension	Hyperglycemia / Insulin resistance ⁴
Hyponatremia ¹ (important AE)	Hyperlipidemia ⁵
Hypokalemic metabolic alkalosis	GI intolerance (rare)
Hypomagnesemia ²	Pregnancy: May decrease placental perfusion (caution)
Hypercalcemia (uncommon)	

Mechanisms:

1. NCC blockade impairs urine dilution → impairs free water excretion. Concentrating ability is not impaired so collecting duct responds to ADH release → AQP2 insertion → free water reabsorption + thirst (water consumption) → dilutional hyponatremia. (Recall: TAL / early DCT form the diluting segment.)

Concomitant NSAID increases risk by enhancing ADH effect (PGs normally counteract ADH.)

2. Chronic thiazide use leads to DCT remodeling and downregulation of apical TRPM6 magnesium channel, which decreases Mg^{2+} reabsorption / increases magnesium wasting.

3. Hypovolemia associated enhanced uric acid reabsorption and reduced secretion via URAT1 (an OAT)

4. Hypokalemia impairs insulin release and thiazides reduce insulin sensitivity → insulin resistance

5. Mild alterations in lipid metabolism related to impaired insulin sensitivity.

Check your knowledge

How do thiazide diuretics reduce blood pressure for the management of hypertension?

What is the mechanism for the use of thiazide diuretics in the treatment of vasopressin resistance (AVP-R also known as nephrogenic diabetes insipidus), including lithium-induced AVP-R?

What is the mechanism for the thiazide diuretics' calcium preserving effects?

Check your knowledge

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What is the mechanism of potassium-wasting and metabolic alkalosis caused by thiazide diuretics?

What is the mechanism for thiazide-induced hyponatremia?

Check your knowledge

What is the mechanism for thiazide-induced insulin resistance?

How does thiazide diuretic therapy increase the risk of lithium toxicity in patients taking lithium?

How does thiazide diuretic therapy increase the risk of digoxin toxicity?

What is the effect of concomitant probenecid on thiazide diuretic therapy?

ENaC Inhibitors Clinical Uses

Hypertension	ENaC inhibitor + thiazide diuretic combination therapy to: Enhance diuresis Counteract hypokalemia induced by thiazide diuretics
Edema	
To reduce lithium-induced vasopressin resistance in patients who need to continue lithium therapy	1. Amiloride prevents ENaC-mediated transport of Li^+ into principal cells → reduces Li^+ accumulation and AQP2 downregulation 2. ENaC + thiazide → Additive effect: Thiazide → mild ECF contraction that increases proximal Na^+/water reabsorption → decreases distal Na^+/water delivery → lowers urine volume → reduces polyuria. <i>Carefully monitor lithium levels.</i>
Pathophysiology of lithium-induced vasopressin resistance (nephrogenic diabetes insipidus): Lithium enters principal cells via ENaC and is not transported out. Intracellular accumulation due to chronic lithium exposure leads to principal cell apoptosis, downregulation of AQP2 expression, and collecting duct remodeling. Lithium-induced AVP-R may be irreversible if tubular injury is severe.	
Liddle syndrome (pseudoaldosteronism)	ENaC inhibitor + sodium restriction for treatment of hypertension and hypokalemia caused by the disorder

Liddle syndrome is a rare autosomal dominant disorder resulting from a gain of function mutation of genes that regulate ENaC. It affects the C-terminal portion of the β or γ subunit of the ENaC, resulting in unregulated reabsorption of Na^+ and secretion of K^+ and H^+ . Plasma renin and aldosterone are suppressed due to Na^+ and fluid retention.

ENaC Inhibitors: Cautions

Patient factors that increase the risk of hyPERkalemia

- Older age
- High-dose ENaC inhibitor therapy
- Renal impairment
- Hypoaldosteronism
- Combination with other drugs that impair renal K⁺ excretion

Drug Interactions that increase the risk of hyPERkalemia

- ACEs, ARBs, DRIs
RAS inhibitors ↓ aldosterone secretion which causes ↑ K⁺ reabsorption
- **NSAIDs**: suppression of PG-dependent renin release leads to ↓ ANG II-mediated aldosterone secretion → ↑ K⁺ reabsorption
- K⁺ supplements
- Calcineurin inhibitors (additive hyperkalemic effects)

K⁺-sparing diuretics are ***contraindicated*** in patients with hyperkalemia or at increased risk of developing hyperkalemia.

ACE: angiotensin converting enzyme inhibitor
ARB: angiotensin receptor blocker
DRI: direct renin inhibitor
RAS: renin-angiotensin system

Spironolactone and Eplerenone: Clinical Uses

Edema	MR antagonist is usually combined with a thiazide or loop diuretic for its potassium-sparing effect and added diuretic effect.
Hypertension	
Primary aldosteronism: Idiopathic adrenal hyperplasia (bilateral)	Goals of therapy: To raise serum K ⁺ to high-normal range, normalize blood pressure, reverse effects of hyperaldosteronism on heart and kidneys
Heart failure HFrEF and HFpEF patients receiving optimal therapy and low risk for hyperkalemia	MR overexpression in nonepithelial tissues, such as blood vessels and heart, contribute to inflammation, myocardial remodeling and fibrosis, in addition to sodium retention and potassium loss at the distal tubules.
Liver cirrhosis Splanchnic vasodilation causes low arterial volume, which activates RAAS, promoting Na ⁺ /water retention, worsening ascites.	Spironolactone (often given with furosemide) Diuretic of choice – long-acting and does not cause hypokalemia Hypokalemia increases ammonia levels by increasing proximal tubule uptake of glutamine (forms ammonia) and upregulation of enzymes involved in ammonia production.
Acne, hirsutism in females Transgender women	Spironolactone: Antiandrogenic effects

Check your knowledge

Based on their mechanisms of action, why are ENaC inhibitors and MR antagonists referred to as weak diuretics?

How do the mechanisms of action of the ENaC inhibitors and MR antagonists differ?

What is the mechanism for the potassium-sparing effect of ENaC inhibitors and MR antagonists?

Check your knowledge

What is the role of potassium-sparing diuretics in the management of hypertension?

What other drugs increase the risk of hyperkalemia when administered with an ENaC inhibitor or an MR antagonist?

Why is an MR antagonist the recommended drug of choice for treating ascites due to hepatic cirrhosis?

Check your knowledge

What is the role of ENaC inhibitors in the treatment of lithium-induced AVP-R (nephrogenic diabetes insipidus)?

Why is an ENaC inhibitor a recommended treatment in Liddle syndrome?

Why is an MR antagonist recommended in HFrEF and HFpEF in patients stabilized on standard therapy?

What are the mechanistic and clinical rationales for treating acne in females with spironolactone?

A 70-year-old woman with hypertension is treated with hydrochlorothiazide and spironolactone. After beginning daily naproxen for knee pain, her labs show rising creatinine and potassium level 5.8 mEq/L (dangerously high).

Explain the mechanism of the drug-drug interaction leading to the adverse drug event.

Osmotic Diuretics: Adverse Effects / Cautions

ECF volume expansion:

Common side effects: Headache, dizziness, nausea, vomiting (common)

Heart failure complication: Florid pulmonary edema as fluid overwhelms the heart's ability to pump it out, especially patients with left-sided heart failure, leading to backup into the lungs, pressure on the pulmonary capillaries, and leakage into alveolar spaces

Hyponatremia: In patients with severe renal impairment, mannitol cannot be excreted → retained in the blood → osmotic extraction of water from cells into vasculature → hyponatremia

Dehydration and electrolyte imbalance:

Rapid diuresis → loss of water in excess of electrolytes → dehydration and hypovolemia → hypernatremia and hyperkalemia

Profound diuresis (excessive dosing) may cause fluid and electrolyte loss.

Nephrotoxicity: Acute kidney injury (risk factors: pre-existing kidney dysfunction, dose and rate of administration)

Incidence estimated at 6%-7% of patients who receive the drug

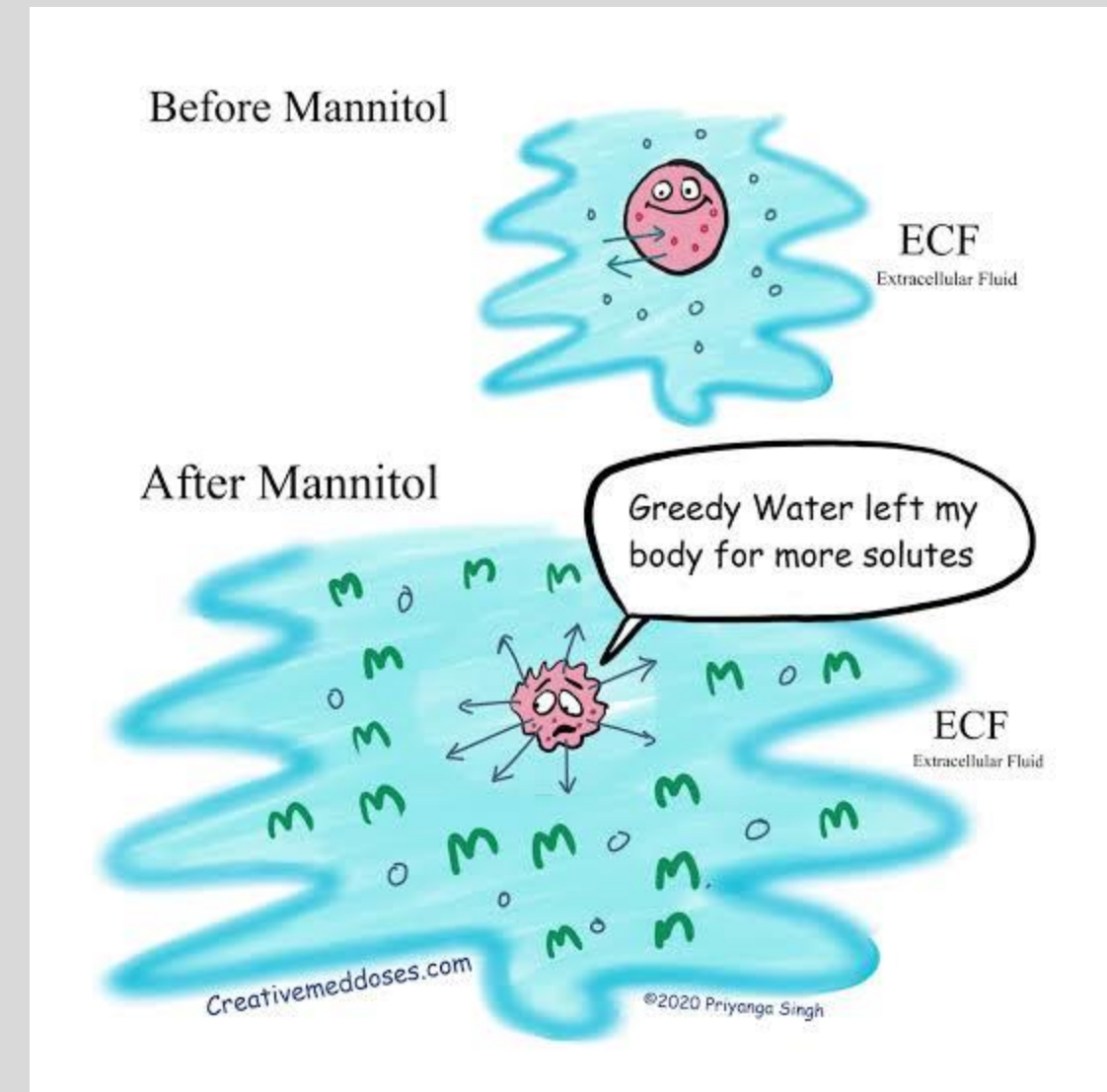
Monitor. Correct electrolyte disturbances. Avoid dehydration.
Discontinue if cardiac or pulmonary status worsens or CNS toxicity develops.

Check your knowledge

What is the primary mechanism of action of mannitol?

Which parts of the nephron is primarily affected by mannitol?

What is mannitol's primary therapeutic use?



Check your knowledge

What is a potential serious adverse effect of mannitol?

Why is mannitol contraindicated in patients with anuria?

What are the pharmacokinetic properties of mannitol used as a diuretic agent?

A 65-year-old male is admitted for treatment of a traumatic brain injury and elevated intracranial pressure. He is treated with intravenous mannitol, resulting in initial improvement in neurologic status. Several hours later, he develops acute shortness of breath, crackles on lung exam, and decreased oxygen saturation. Chest imaging reveals bilateral pulmonary edema.

What is the mechanism of this complication?

What risk factors for this complication?

Volume contraction and compensatory responses to diuretic therapy

- Volume depletion activates RAAS and reduces GFR.
- RAAS activation $\uparrow$ aldosterone and $\uparrow$ distal Na^+ reabsorption.
- ADH (vasopressin) release causes AQP2 insertion in collecting duct and free water reabsorption.
- Distal nephron compensation limits natriuresis and drives K^+ and H^+ loss.
- *Monitor for hypotension, prerenal azotemia, and renal dysfunction.*

Therapeutic uses

- Edema/CHF: loop diuretics reduce ECF; diuretic resistance potential; loop + thiazide synergy by sequential nephron block.
- Hypertension: thiazides first-line; add ENaC inhibitor or MRA to moderate potassium-wasting effect and increase natriuretic/diuretic effect
- Hypercalcemia: loop diuretics as adjunct after IV hydration – more effective treatments are preferred
- Calcium stones with hypercalciuria: thiazides (selected patients), lifestyle and dietary modifications, weight loss.
- Metabolic alkalosis, mountain sickness, glaucoma: acetazolamide.
- Cirrhosis/ascites: spironolactone first-line – management of hyperaldosteronism.

Drug interactions and clinical cautions

- Thiazides $\uparrow$ lithium levels ($\uparrow$ proximal Na^+/Li^+ reabsorption).
- Loops have a lower risk of lithium toxicity but still require monitoring.
- ACEi/ARBs + K^+ -sparing $\rightarrow$ hyperkalemia risk.
- NSAIDs increase risk of acute kidney injury, worsen edema and heart failure symptoms, increase risk of hyponatremia (thiazides), weaken BP control, blunt diuretic efficacy, worsen metabolic alkalosis.

NSAIDs-Diuretics Interactions and Effects

NSAIDs worsen renal hemodynamics, reduce diuretic delivery, promote sodium retention, and impair potassium excretion.

Diuretic Class	Mechanism	Resulting Adverse Effect	Why it happens
Thiazides	Block PG-mediated afferent vasodilation	↓GFR Increase AKI risk	Thiazide-induced volume depletion → kidney becomes PG-dependent
	Decrease free water clearance	Hyponatremia	NSAID removes ADH inhibition + thiazide impairs urine dilution
	Decrease diuretic delivery	Effect on BP, edema is blunted	Afferent constriction → ↓RBF reduces GFR → less drug is delivered
Loop diuretics	Afferent constriction → ↓RBF	↓natriuresis diuretic resistance	Afferent constriction → ↓RBF reduces GFR → less drug is delivered
	↑RAAS, ↓PGs	Worsen metabolic alkalosis	Loss of PG-mediate afferent dilation → perceived as low perfusion → RAAS activation → ↑K ⁺ , H ⁺ secretion, bicarbonate reabsorption
	Sodium retention	Worsen edema, heart failure	NSAIDs promote Na ⁺ retention in TAL/CD
K-sparing diuretics	↓GFR → ↓K ⁺ excretion	Hyperkalemia	NSAIDs and drug ↓K ⁺ excretion – additive
ACEIs / ARBs (RAS inhibitors)	Afferent constriction Efferent dilation	AKI triple hit	NSAID + ACEI/ARB + diuretic → collapse of glomerular filtration pressure
	↓aldosterone, ↓GFR	Severe hyperkalemia	NSAID + RAS inhibitor both ↓K ⁺ secretion

How do thiazide diuretics reduce blood pressure for the management of hypertension?

- Acutely lower BP by reducing ECF and cardiac output.
- Sustained effect is due to chronic sodium depletion that decreases vascular smooth muscle responsiveness to vasoconstrictors → sustained reduction in peripheral vascular resistance.

What is the mechanism for the use of thiazide diuretics in the treatment of vasopressin resistance (AVP-R also known as nephrogenic diabetes insipidus), including lithium-induced AVP-R?

- Thiazide-induced mild volume contraction → RAAS activation and ↓GFR → Na^+ and water reabsorption in proximal tubule → substantial drop in filtrate reaching ADH-sensitive sites in collecting ducts → overall urine output decreases by up to 30-50% → new lower steady state is established. Urine output remains chronically lower than baseline (as long as the drug is taken).
- Treatment of AVP-R, including lithium-induced AVP-R: Discontinue lithium (if the patient is taking it) + thiazide diuretic + low sodium, low protein diet + NSAID.

What is the mechanism for the thiazide diuretics' calcium preserving effects?

- Blocking luminal NCC in the DCT lowers intracellular Na^+ concentration → enhances basolateral NCX1 activity (3Na^+ in, 1Ca^{2+} out), which creates stronger gradient for Ca^{2+} entry via TRPV5 → ↑ Ca^{2+} into the cell → exported out via basolateral NCX1. PMCA1b (ATPase) fine-tunes Ca^{2+} export (a secondary mechanism).

What is the mechanism of the potassium-wasting and hypokalemic metabolic alkalosis caused by thiazide diuretics?

- Blocking NCC in the DCT increases sodium and water delivery to the collecting ducts.
- Delivery of an increased sodium load to the aldosterone-sensitive sites in the distal nephron along with volume contraction → aldosterone (due to RAAS activation) upregulates ENaC and Na⁺/K⁺ ATPase → increases sodium reabsorption by ENaC → enhances the lumen negative potential → increases K⁺ excretion → potassium-wasting effect
- K⁺ loss / hypokalemia, increased H⁺ secretion, ECF volume contraction / RAAS activation → hypokalemic metabolic alkalosis. Hypochloremia and bicarbonate retention sustain metabolic alkalosis.

What is the mechanism for thiazide-induced hyponatremia?

Thiazide diuretics impair the kidney's ability to dilute urine, while concentrating ability remains intact. **Retention of water is greater than retention of sodium → dilutional hyponatremia.**

- (1) **NCC inhibition impairs urine dilution:** Free water is generated in the diluting segments by removing solute without water. NCC inhibition decreases NaCl reabsorption, thereby increasing solute concentration, which impairs excretion of free water.
- (2) **Urine concentrating ability is preserved:** The medullary gradient remains intact, and the collecting duct responds to ADH → AQP2 insertion in collecting ducts → free water reabsorption.
- Increased thirst tends to increase fluid intake.

Most patients do not develop hyponatremia unless water excretion is impaired.

Check your knowledge

What is the mechanism for thiazide-induced insulin resistance?

- Hypokalemia disrupts the function of the ATP-sensitive K⁺ channels in pancreatic beta cells, leading to impaired insulin release (primary mechanism) + thiazides reduce tissue utilization of glucose (impaired insulin sensitivity) → hyperglycemia.

How does thiazide diuretic therapy increase the risk of lithium toxicity in patients taking lithium?

- Blocking NCC increases Na⁺ excretion → compensatory reabsorption of both Na⁺ and Li⁺ in the proximal tubule → ↑ plasma levels of lithium (narrow therapeutic window) → even a modest increase in serum lithium levels can lead to lithium toxicity.

How does thiazide diuretic therapy increase the risk of digoxin toxicity?

- Hypokalemia augments digitalis-mediated increase in cytosolic calcium levels → intracellular Ca²⁺ overload → increased cardiac automaticity → arrhythmia
- FYI: Digoxin competes with K⁺ binding to the Na⁺/K⁺ ATPase in cardiomyocytes. When K⁺ is low (hypokalemia), more digoxin binds → stronger inhibition → digoxin toxicity

What is the effect of concomitant probenecid on thiazide diuretic therapy?

- Probenecid is a potent inhibitor of organic anion transporters (OATs). Diuretics, including thiazide diuretics enter the nephron predominantly via OATs. Blocking the OATs reduces the concentration of diuretic in the nephron, which can reduce diuretic efficacy.

Check your knowledge

Based on their mechanisms of action, why are ENaC inhibitors and MR antagonists referred to as weak diuretics?

- Approximately 2-5% of sodium excretion occurs in the principal cells. Therefore, ENaC inhibitors and MR antagonists block only a small fraction of the total sodium, thus water, reabsorption by the kidneys.

How do the mechanisms of action of the ENaC inhibitors and MR antagonists differ?

- ENaC inhibitors block sodium reabsorption through the luminal epithelial sodium channels.
- MR antagonists pass through the basolateral membrane and block the mineralocorticoid receptor in the cytosol. The complex translocates to the nucleus, binds specific response elements on the DNA, which stimulates transcription and expression of various genes involved in the sodium reabsorption and potassium excretion.
- Blocking either site of action reduces Na^+ reabsorption / increases Na^+ excretion, followed by water.

What is the mechanism for the potassium-sparing effect of ENaC inhibitors and MR antagonists?

- Blocking Na^+ reabsorption in the principal cells abolishes the transepithelial potential difference, which reduces the electrochemical gradient for potassium. K^+ is retained, which can lead to hyperkalemia.

Check your knowledge

What is the role of potassium-sparing diuretics in the management of hypertension?

- ENaC inhibitors and MR antagonists are frequently combined with a thiazide diuretic to counteract the potassium-wasting effect of the thiazide diuretic and for added diuretic effect.

What other drugs increase the risk of hyperkalemia when administered with an ENaC inhibitor or an MR antagonist?

- RAS inhibitors (ACE inhibitors, ARBS, and direct renin inhibitors), NSAIDs, potassium supplements, and calcineurin inhibitors

Why is an MR antagonist the recommended drug of choice for treating ascites due to hepatic cirrhosis?

- Spironolactone is drug of choice in the treatment because of its aldosterone antagonism effects.

Mechanism:

- Portal hypertension increases hydrostatic pressure, which leads to fluid transudation into the peritoneal cavity, which contributes to effective hypovolemia.
- Splanchnic vasodilation is triggered by nitric oxide and other vasodilators in response to portal hypertension, which leads to reduced systemic vascular resistance, which activates SNS and RAAS and vasopressin leading to sodium and water retention.
- Hypoalbuminemia due to liver dysfunction decreases oncotic pressure, which facilitates fluid leakage from the capillaries into the interstitial and peritoneal spaces.

Check your knowledge

What is the role of ENaC inhibitors in the treatment of lithium-induced AVP-R (nephrogenic diabetes insipidus)?

- ENaC inhibition counteracts vasopressin resistance by preventing Li^+ entry to principal cells.
- Lithium enters principal cells via ENaC and is not pumped out efficiently. Li^+ accumulation in the principal cells interferes with expression and trafficking of AQP2. Thus, the collecting duct becomes impermeable to water. ENaC inhibition prevents Li^+ entry to principal cells.

Why is an ENaC inhibitor a recommended treatment in Liddle syndrome?

- Liddle syndrome is a rare autosomal dominant disorder resulting from a gain of function mutation of genes that regulate ENaC.
- ENaC inhibitor + sodium restriction for treatment of hypertension and hypokalemia caused by the disorder

Why is an MR antagonist recommended in HFrEF and HFpEF in patients stabilized on standard therapy?

- Evidence supports the use of spironolactone in conjunction with standard therapy in patients at low risk for hyperkalemia by blocking the effects of aldosterone overexpression.

What are the mechanistic and clinical rationales for treating acne in females with spironolactone?

- Spironolactone has antiandrogen effects and shifts estrogen/testosterone balance toward estrogen.

A 70-year-old woman with hypertension is treated with hydrochlorothiazide and spironolactone. After beginning daily naproxen for knee pain, her labs show rising creatinine and potassium level 5.8 mEq/L (dangerously high).

Explain the mechanism of the drug-drug interaction leading to the adverse drug event.

Triple hit:

- Thiazides reduce effective circulating volume → ↓ renal perfusion pressure. Therefore, compensatory mechanisms maintain GFR, including prostaglandin-mediate afferent dilation.
- NSAIDs block renal prostaglandin synthesis, leading to afferent arteriolar constriction, which reduces renal blood flow, therefore reduces GFR.
- Thiazide + NSAID together cause a **marked reduction in GFR**. Decreased GFR increases serum creatinine and risk of acute kidney injury.
- Spironolactone decreases ENaC activity and Na⁺ reabsorption, which removes the lumen-negative potential, therefore decrease K⁺ secretion via ROMK. Potassium is retained.
- NSAIDs-induced inhibition or renin release decreases aldosterone-mediated secretion of K⁺ → K⁺ retention

Mechanism	Effect
Thiazide	Volume depletion
NSAID	↓afferent dilation → ↓GFR
Spironolactone	↓K ⁺ secretion
↓ GFR	↓K ⁺ excretion

Check your knowledge

What is the primary mechanism of action of mannitol?

- Mannitol (a sugar alcohol) and other osmotic diuretics increase the concentration of solutes in the plasma and interstitial spaces and in the renal tubules.
- The elevated osmolarity produces an osmotic gradient, which draws water from areas of lower solute concentration, like the brain parenchyma and eye, into the plasma and interstitial fluid and the renal tubules. The water is excreted as urine.

Which parts of the nephron is primarily affected by mannitol?

- The proximal tubule and descending limb of the loop of Henle are permeable to water. Mannitol osmotically draws water into the nephron at these segments. Mannitol also impairs water reabsorption in the collecting tubule.

What is mannitol's primary therapeutic use?

- Intracranial pressure (ICP). By increasing solute concentration in plasma, mannitol osmotically draws water out of the brain parenchyma into the intravascular space, reducing ICP. The water is excreted in the urine.
- Intraocular pressure (IOP): By the same mechanism, mannitol reduces IOP in acute emergencies where rapid IOP reduction is critical such as acute angle-closure glaucoma and optic trauma.

Check your knowledge

What is a potential serious adverse effect of mannitol?

- Volume expansion leading to pulmonary edema in patients with left-side heart failure as fluid overload causes the fluid backs up into the lungs, increasing pulmonary capillary pressure, causing leakage into alveolar spaces.
- Volume expansion causing electrolyte imbalance with hyponatremia due to fluid overload or hypernatremia.
- Dehydration with rapid water loss greater than electrolyte loss leading to hypernatremia and hyperkalemia.
- Profound diuresis causing fluid and electrolyte loss.

Why is mannitol contraindicated in patients with anuria?

- In patients not making urine (anuria), mannitol accumulates in the extracellular fluid, leading to volume overload and pulmonary edema, potentially hypertensive crisis, hypertensive encephalopathy, renal failure.

What are the pharmacokinetic properties of mannitol used as a diuretic agent?

- Mannitol is administered intravenously; dosed to achieve rapidly distributes to extracellular compartment; freely filtered by the glomerulus; minimal reabsorption (stays in the nephron); onset of ICP reduction in about 15 to 30 minutes; duration of ICP reduction 1.5-6 hours.

Check your knowledge

What is the primary mechanism of action of mannitol?

- Mannitol (a sugar alcohol) and other osmotic diuretics increase the concentration of solutes in the plasma and interstitial spaces and in the renal tubules.
- The elevated osmolarity produces an osmotic gradient, which draws water from areas of lower solute concentration, like the brain parenchyma and eye, into the plasma and interstitial fluid and the renal tubules. The water is excreted as urine.

Which parts of the nephron is primarily affected by mannitol?

- The proximal tubule and descending limb of the loop of Henle are permeable to water. Mannitol osmotically draws water into the nephron at these segments. Mannitol also impairs water reabsorption in the collecting tubule.

What is mannitol's primary therapeutic use?

- Intracranial pressure (ICP). By increasing solute concentration in plasma, mannitol osmotically draws water out of the brain parenchyma into the intravascular space, reducing ICP. The water is excreted in the urine.
- Intraocular pressure (IOP): By the same mechanism, mannitol reduces IOP in acute emergencies where rapid IOP reduction is critical such as acute angle-closure glaucoma and optic trauma.

Check your knowledge

What is a potential serious adverse effect of mannitol?

- Volume expansion leading to pulmonary edema in patients with left-side heart failure as fluid overload causes the fluid backs up into the lungs, increasing pulmonary capillary pressure, causing leakage into alveolar spaces.
- Volume expansion causing electrolyte imbalance with hyponatremia due to fluid overload or hypernatremia.
- Dehydration with rapid water loss greater than electrolyte loss leading to hypernatremia and hyperkalemia.
- Profound diuresis causing fluid and electrolyte loss.

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A 65-year-old male is admitted for treatment of a traumatic brain injury and elevated intracranial pressure. He is treated with intravenous mannitol, resulting in initial improvement in neurologic status. Several hours later, he develops acute shortness of breath, crackles on lung exam, and decreased oxygen saturation. Chest imaging reveals bilateral pulmonary edema.

What is the mechanism of this complication?

- Mannitol stays in the extracellular space, which increases plasma osmolality.
- The osmotic gradient pulls water from the intracellular to the interstitial to the intravascular compartment, which causes rapid expansion of plasma volume.
- Increased intravascular volume increases venous return to the heart (increases preload), which increases pulmonary capillary pressure.
- High hydrostatic pressure in the pulmonary capillaries pushes fluid into the lung tissue, causing pulmonary edema.

What risk factors for this complication?

- Patients with heart failure, renal impairment, or existing volume overload cannot handle sudden volume expansion.